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Université: Institut Universitaire des Sciences (IUS)

TD N° 6: Réseau1

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Niveau: 3^{ème} Année

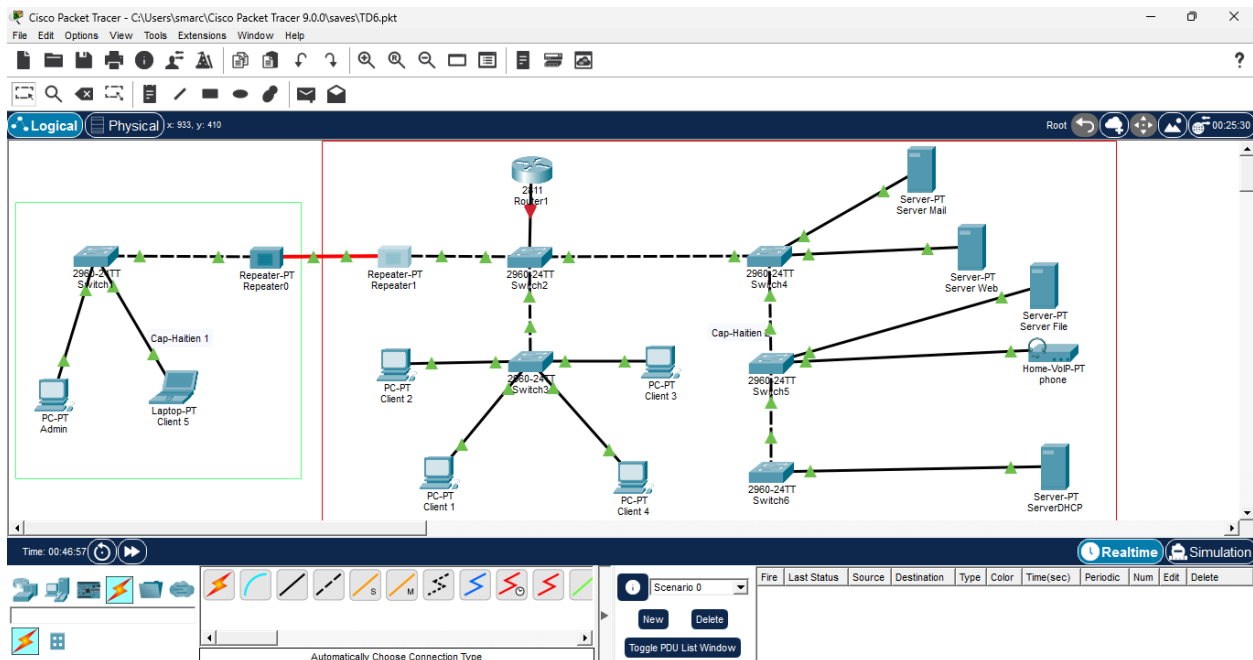
Date: Le 08 /Jan/2026

Objectif

Mettre en place les services DHCP et DHCPv6 afin d'attribuer automatiquement les adresses IP aux hôtes du réseau, en utilisant un serveur, soit un routeur configuré comme serveur DHCP.

1. Reproduisez cette topologie en configurant les services DHCP afin d'attribuer automatiquement les adresses IP aux dispositifs du réseau.

1-1 J'ai reproduit la topologie demandée.



1-2 Configuration du Serveur DHCP

Étapes de configuration

1.2.1 configuration du routeur pour IPv4

Router> enable

Router# configure terminal

Router(config)# interface gigabitethernet 0/0

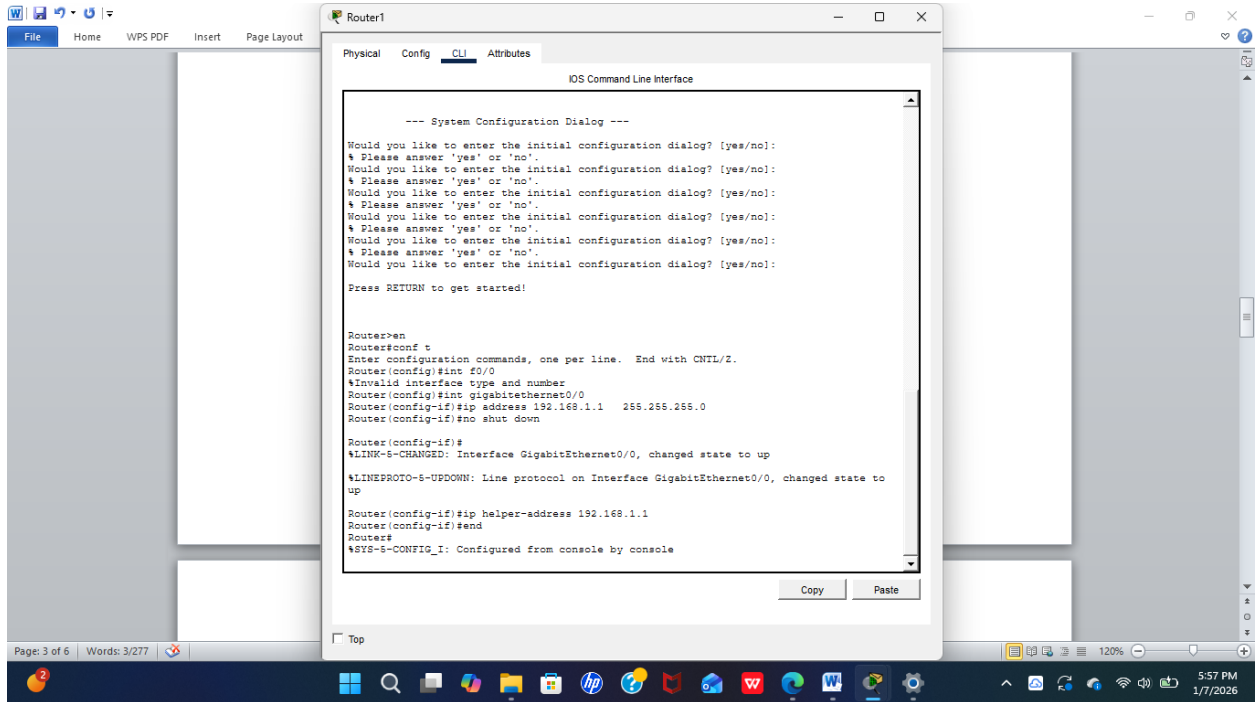
Router(config-if)# ip address 192.168.1.1 255.255.255.0

Router(config-if)# no shutdown

! Indiquer au routeur vers quel serveur DHCP renvoyer les requêtes

Router(config)# ip helper-address 192.168.1.1

Router(config)# end



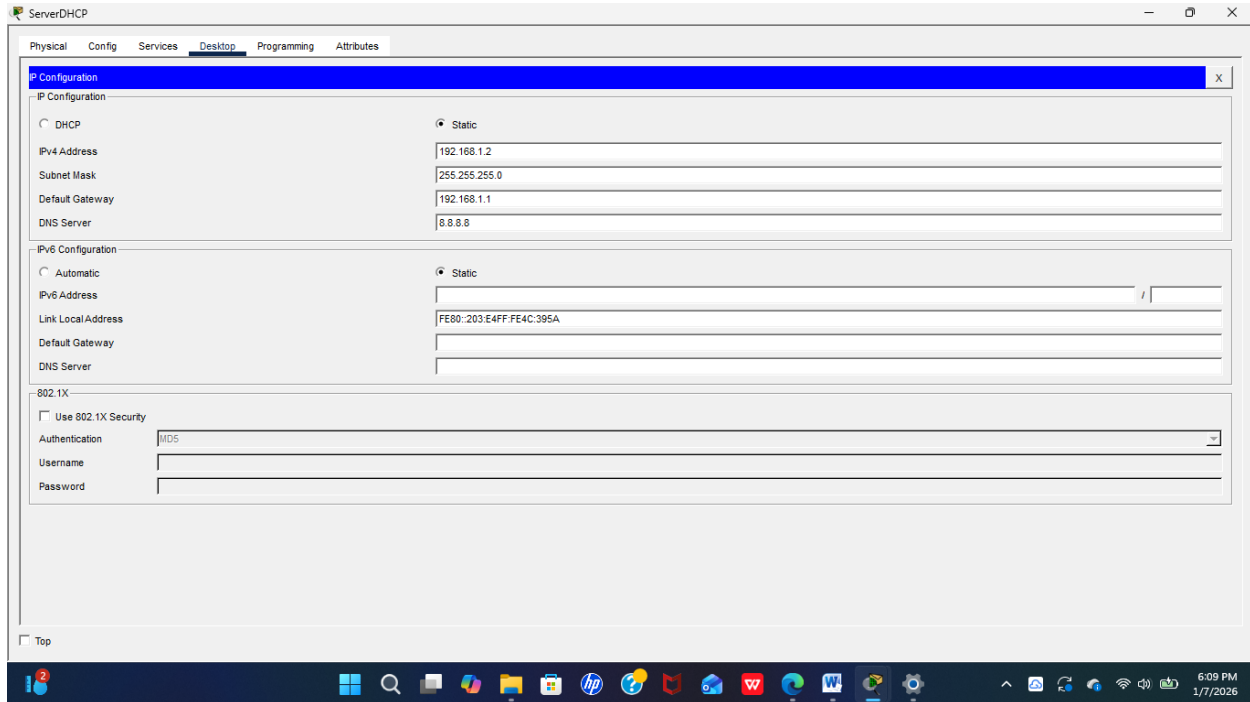
1.2.2 configuration du Serveur DHCP

Onglet Desktop →

IP Configuration : IP : 192.168.1.2

Mask : 255.255.255.0

Gateway : 192.168.1.1



Onglet Services → DHCP : Activer le service DHCP

Créer Pool1 (LAN1) :

Default Gateway : 192.168.1.1

DNS Server : 8.8.8.8

Start IP : 192.168.1.3

Subnet Mask : 255.255.255.0

Max Users : 20

Cisco Packet Tracer - C:\Users\smarc\Cisco Packet Tracer 9.0.0\saves\TD6.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 546, y: 710

29-0-20-17 switch Repeater-PT Repeater0

PC-PT Admin Laptop-PT Client 5

Time: 00:49:18

ServerDHCP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

Start IP Address: 192 168 100 0

Subnet Mask: 255 255 255 0

Maximum Number of Users: 20

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.1.1	0.0.0.0	192.168.100.0	255.255.255.0	20	0.0.0.0	0.0.0.0

Root

Server-PT Server Web

Server-PT Server File

Home-VoIP-PT phone

Server-PT ServerDHCP

Realtime Simulation

Num Edit Delete

6:03 PM 1/7/2026

ServerDHCP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.1.1

DNS Server: 8.8.8.8

Start IP Address: 192 168 1 3

Subnet Mask: 255 255 255 0

Maximum Number of Users: 20

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.1.1	8.8.8.8	192.168.1.3	255.255.255.0	20	0.0.0.0	0.0.0.0

Top

6:06 PM 1/7/2026

Exemple de DHCP4 sur client 1

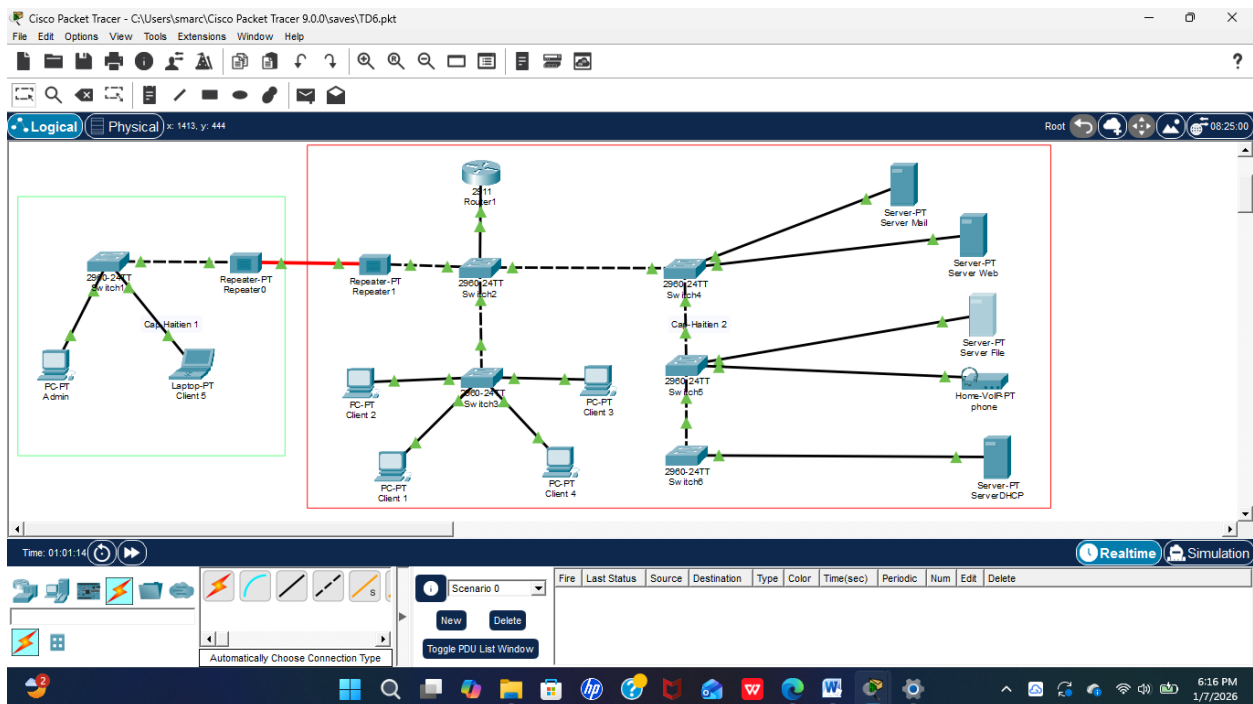
This screenshot shows the Cisco Packet Tracer interface with three main windows. The left window displays a network diagram of 'Cap-Haïtien 1' featuring a switch connected to a PC (Admin) and a laptop (Client 5), with a repeater connected to the switch. The middle window is the configuration page for 'Client 1', showing the 'FastEthernet0' interface with DHCP settings: IPv4 Address 192.168.1.3, Subnet Mask 255.255.255.0, Default Gateway 192.168.1.1, and DNS Server 8.8.8.8. The IPv6 configuration is set to static with a Link Local Address of FE80::202:16FF:FECD:B956. The right window shows a server rack with various services including Mail, Web, File, VoIP phone, and DHCP. The bottom status bar indicates the time is 00:58:15.

Sur client 2

This screenshot shows the Cisco Packet Tracer interface with three main windows. The left window displays the same network diagram as the first image. The middle window is the configuration page for 'Client 2', showing the 'FastEthernet0' interface with DHCP settings: IPv4 Address 192.168.1.4, Subnet Mask 255.255.255.0, Default Gateway 192.168.1.1, and DNS Server 8.8.8.8. The IPv6 configuration is set to static with a Link Local Address of FE80::20A:F3FF:FE19:2159. The right window shows the same server rack configuration. The bottom status bar indicates the time is 00:57:12.

2- Reproduisez cette topologie en configurant les services DHCPv6 afin d'attribuer automatiquement les adresses IP aux dispositifs du réseau.

2.1 la topologie reproduite



2.2 Configuration du Serveur DHCPv6

2.2.1 configuration du routeur

R1> enable

R1# configure terminal

R1(config)# ipv6 unicast-routing

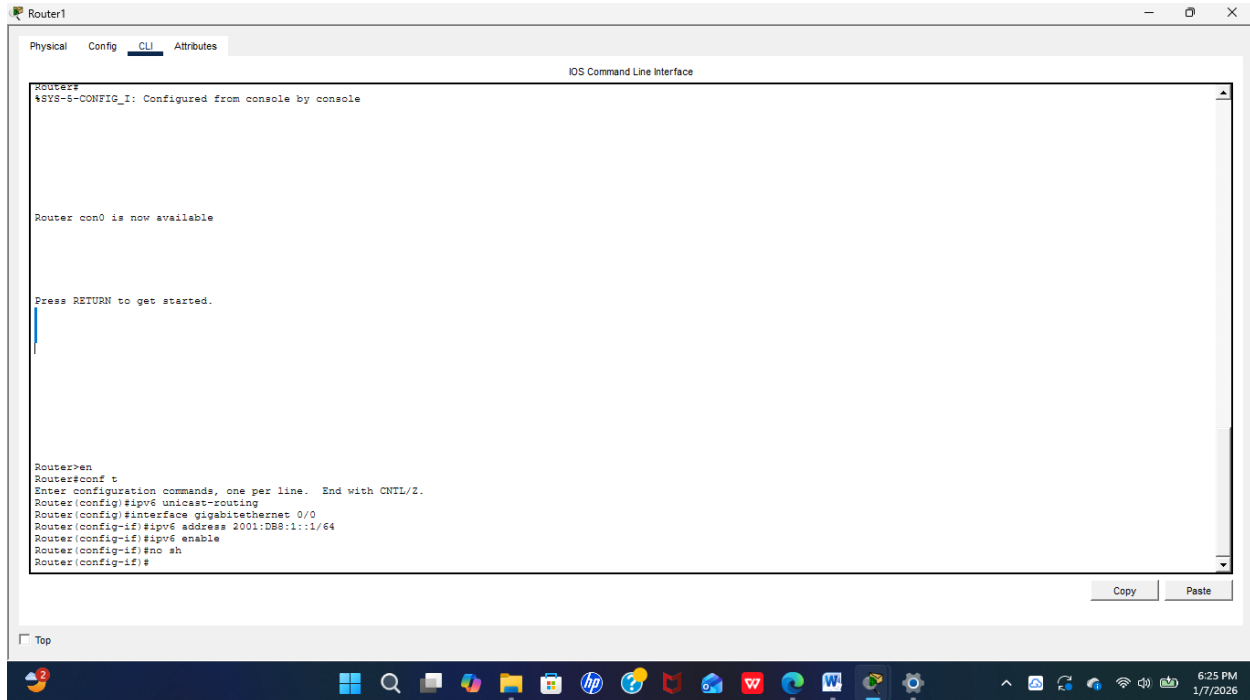
! Interface Cap 1

R1(config)# interface gigabitethernet 0/0

R1(config-if)# ipv6 address 2001:DB8:1::1/64

R1(config-if)# ipv6 enable

R1(config-if)# no shutdown



2.2.2 Configuration du Serveur DHCPv6

Aller dans Desktop → IP Configuration :

IPv6 Address : 2001:DB8:1::2

Prefix : /64

Gateway : 2001:DB8:1::1

Aller dans Services → DHCPv6 :

Activer le service DHCPv6 (On)

Ajouter Pool1 :

Prefix : 2001:DB8:1::/64

DNS Server : 2001:4860:4860::8888

Domain Name : reseau-lan1.com

ServerDHCP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DHCPv6

Interface: FastEthernet0 Service: ☒ On ☐ Off

DHCPv6 Pool: 1

Pool List: 1 Create Pool Remove Pool

DNS Server: 001:4860:4860:8888 Domain Name: reseau-lan1.com

IPv6 Address Prefix

Prefix	Prefix Length	Valid Lifetime	Preferred Lifetime	
2001:DB8:1::3	64	2592000	604800	Create Edit Remove

IPv6 Prefix-Delegation

Prefix	DUID	Local Pool	Valid Lifetime	Preferred Lifetime	
--------	------	------------	----------------	--------------------	--

IPv6 Local Pool

ServerDHCP

☐ Automatic ☒ Static

IPv6 Address: 2001:DB8:1::2 / 64

Link Local Address: FE80::203:E4FF:FE4C:395A

Default Gateway: 2001:DB8:1::1

DNS Server: 2001:4860:4860:8888

802.1X

☐ Use 802.1X Security

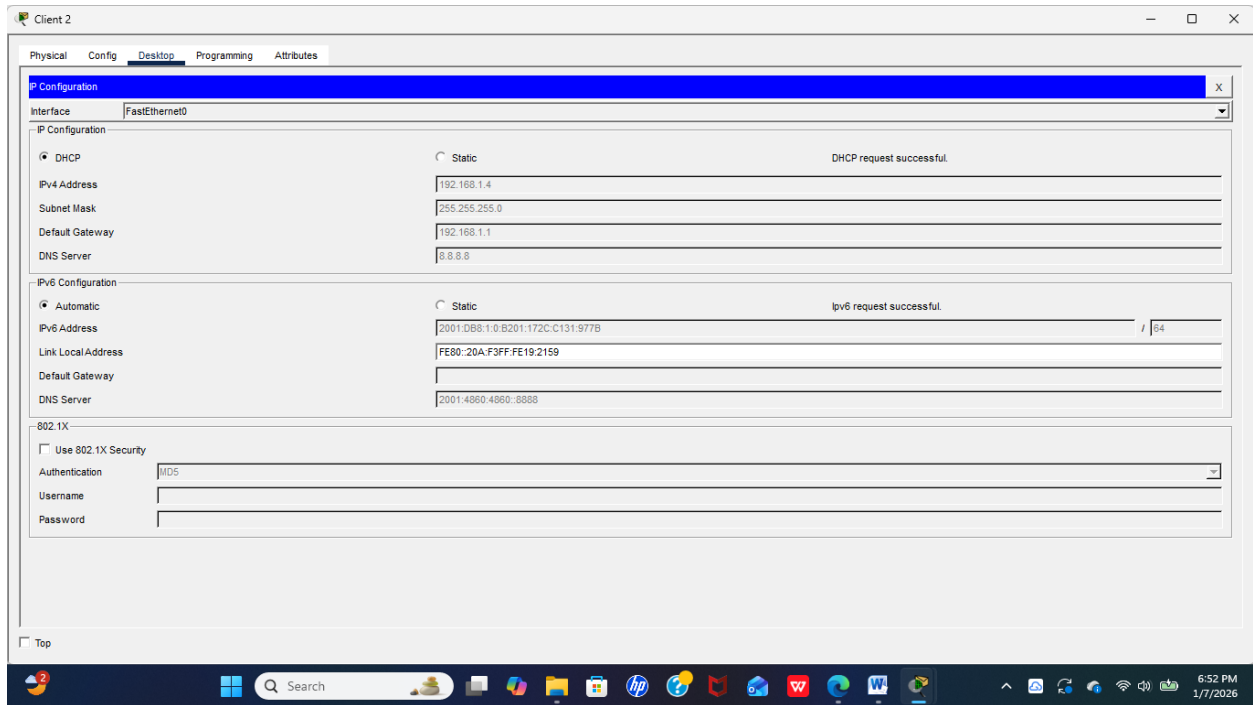
Authentication: MD5

Username:

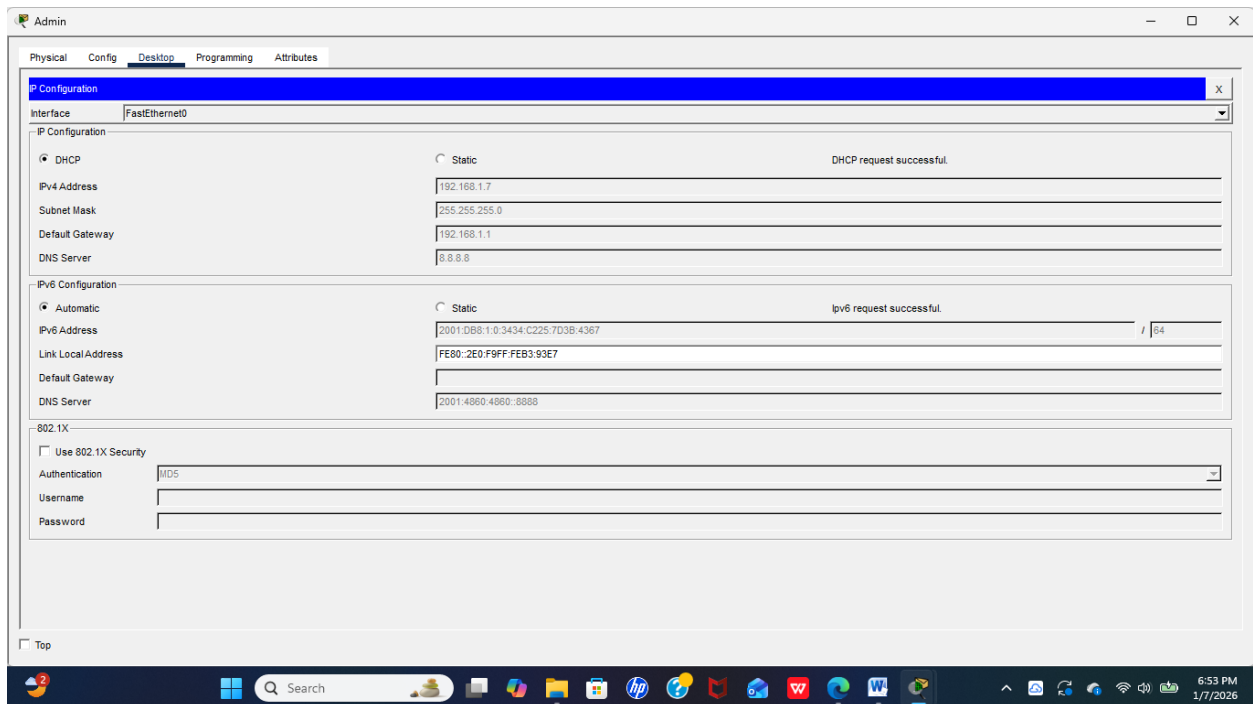
Password:

☐ Top

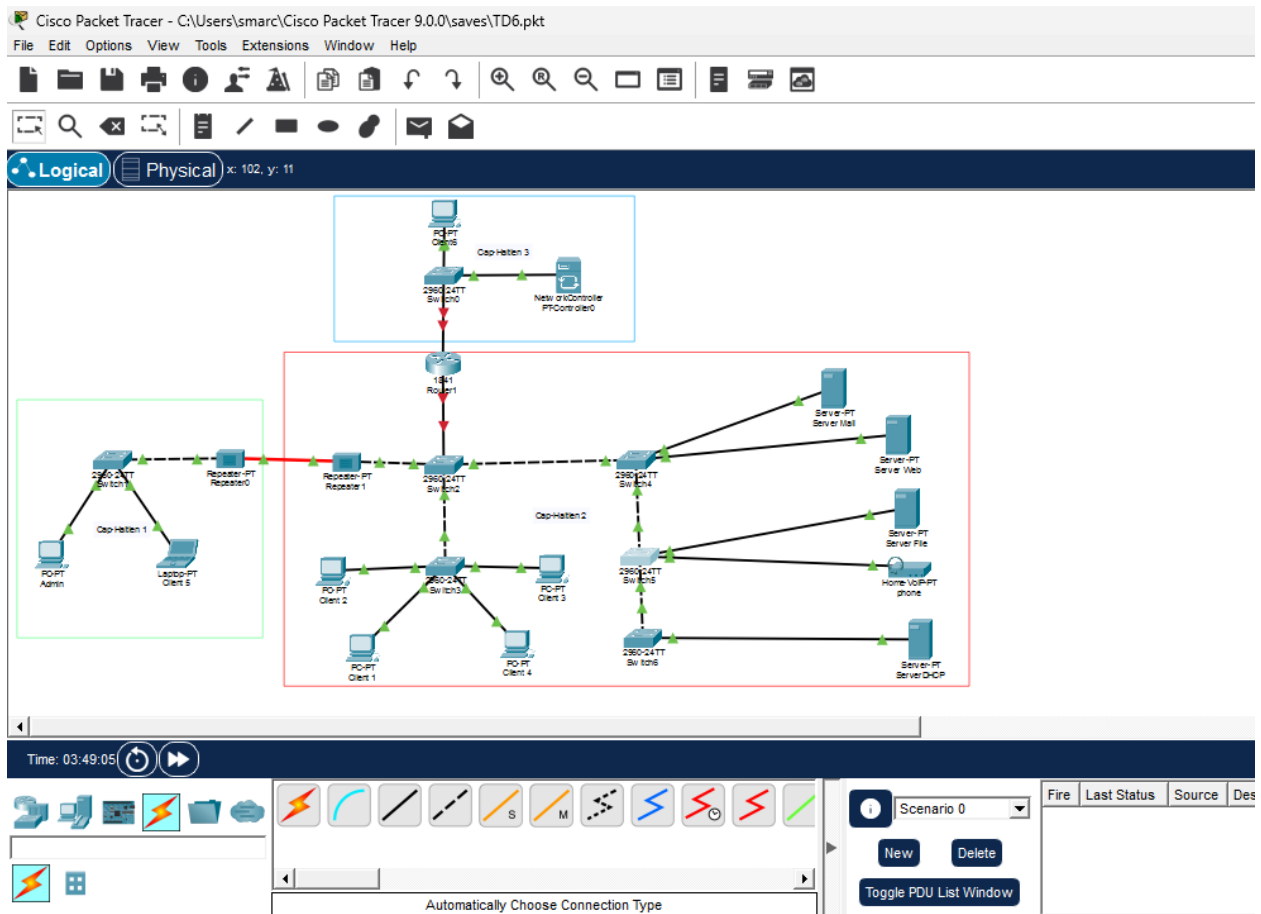
Exemple DHCPv6 sur client 2



Sur admin



3.1 Création de la topologie



3.2 configuration du service DHCP Sur le routeur

3.2.1 configuration des interface du routeur

R1> enable

R1# configure terminal

R1(config)# hostname R1

R1 ! Interface Cap1

R1(config)# interface gigabitEthernet 0/0

R1(config-if)# ip address 192.168.1.1 255.255.255.0

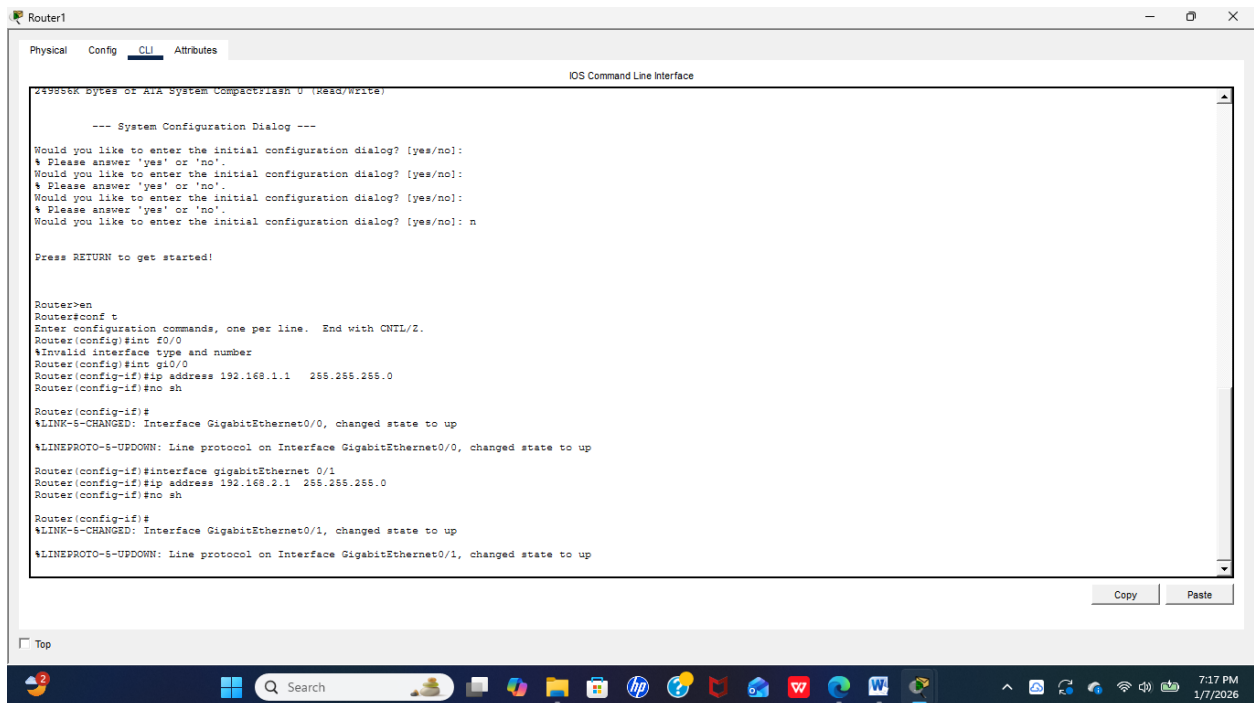
R1(config-if)# no shutdown

Interface Cap2

R1(config)# interface gigabitEthernet 0/1

R1(config-if)# ip address 192.168.2.1 255.255.255.0

R1(config-if)# no shutdown



The screenshot shows a Cisco Router CLI window titled "Router1". The window has tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, showing the "IOS Command Line Interface". The output of the commands is as follows:

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1 ! Interface Cap1
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
Interface Cap2
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown
```

The output also includes system messages such as "Would you like to enter the initial configuration dialog? [yes/no]: n" and "Press RETURN to get started!". At the bottom of the window, there are "Copy" and "Paste" buttons. The Windows taskbar is visible at the bottom of the screen, showing the time as 7:17 PM on 1/7/2026.

3.2.2 Configuration du service DHCP sur le routeur

Pool DHCP pour CAP 1

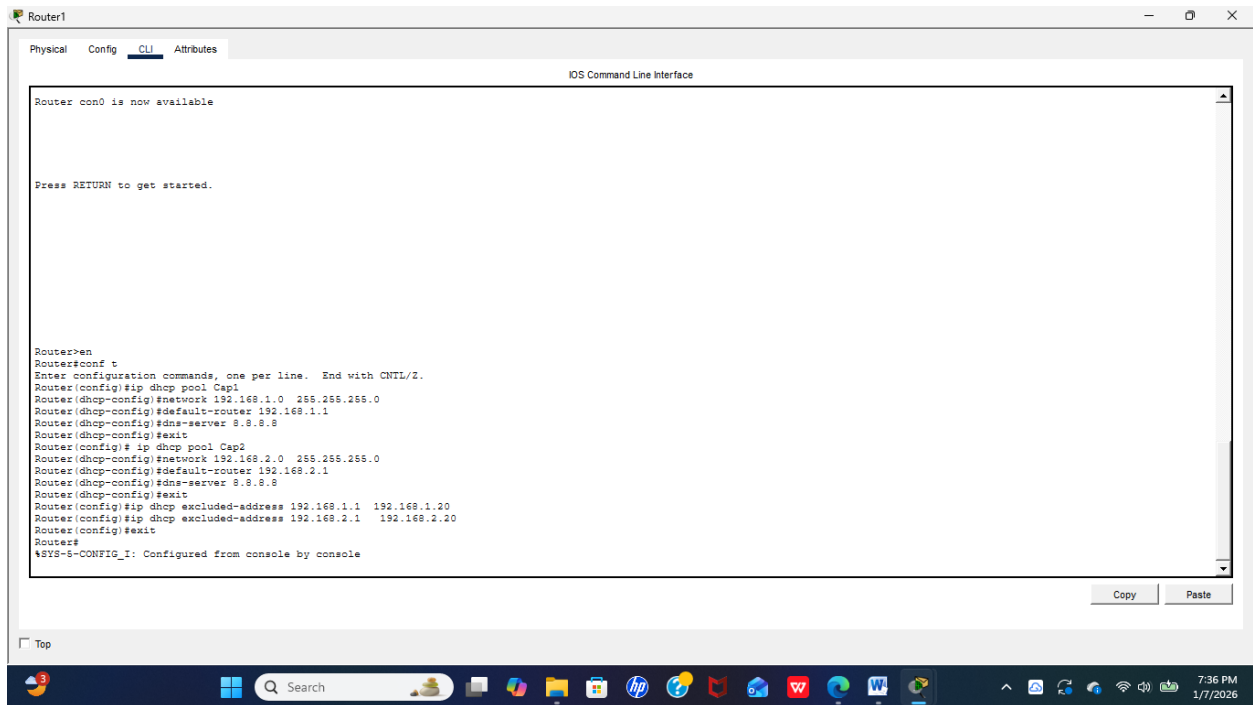
```
R1(config)# ip dhcp pool Cap1
R1(dhcp-config)# network 192.168.1.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.1.1
R1(dhcp-config)# dns-server 8.8.8.8
R1(dhcp-config)# exit
```

! Pool DHCP pour Cap2

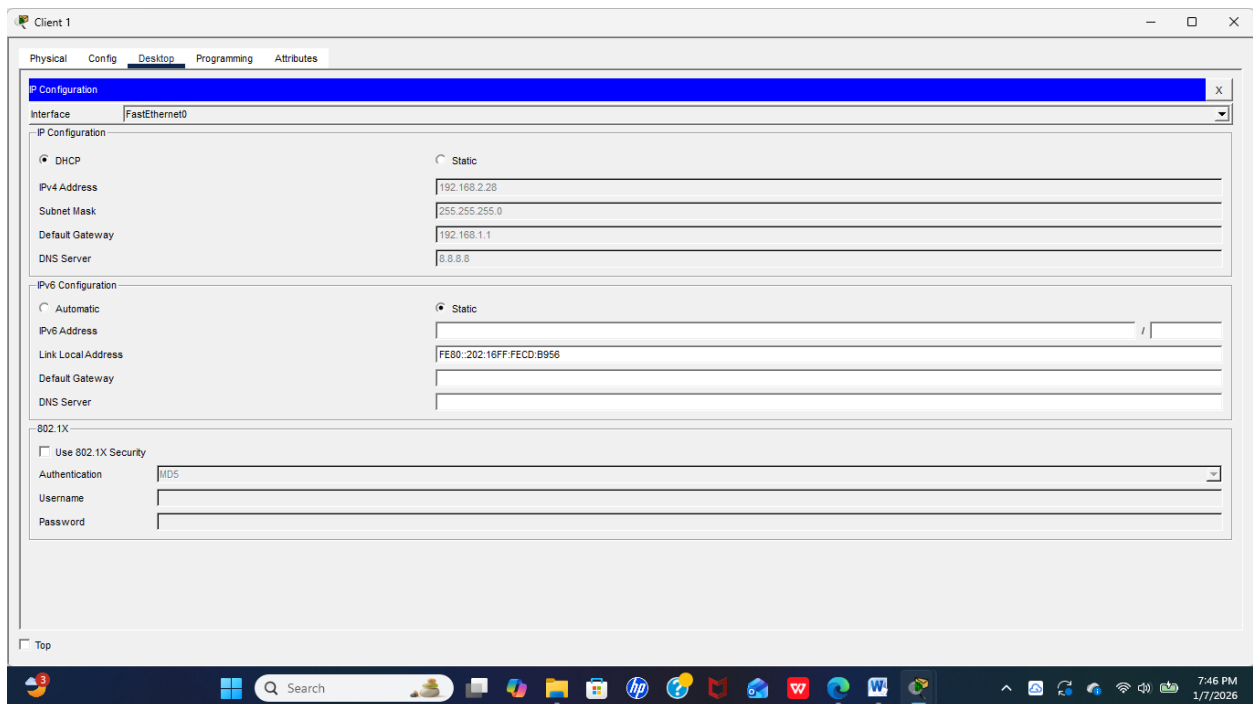
```
R1(config)# ip dhcp pool Cap2
R1(dhcp-config)# network 192.168.2.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.2.1
R1(dhcp-config)# dns-server 8.8.8.8
R1(dhcp-config)# exit
```

! Exclure les adresses réservées (passerelles, serveurs...)

```
R1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.20
R1(config)# ip dhcp excluded-address 192.168.2.1 192.168.2.20
```



Exemple sur client 1 de Cap 2



Cap1, Client 6

Client6

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful

IPv4 Address: 192.168.1.21

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 8.8.8.8

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::250:FFF:FEB3:41C0

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

Top

7:38 PM 1/7/2026

Serveur web

Server Web

Physical Config **Services Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful

IPv4 Address: 192.168.2.29

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

DNS Server: 8.8.8.8

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::210:11FF:FE17:9D72

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

Top

7:49 PM 1/7/2026

3.2.3 Test de connectivité

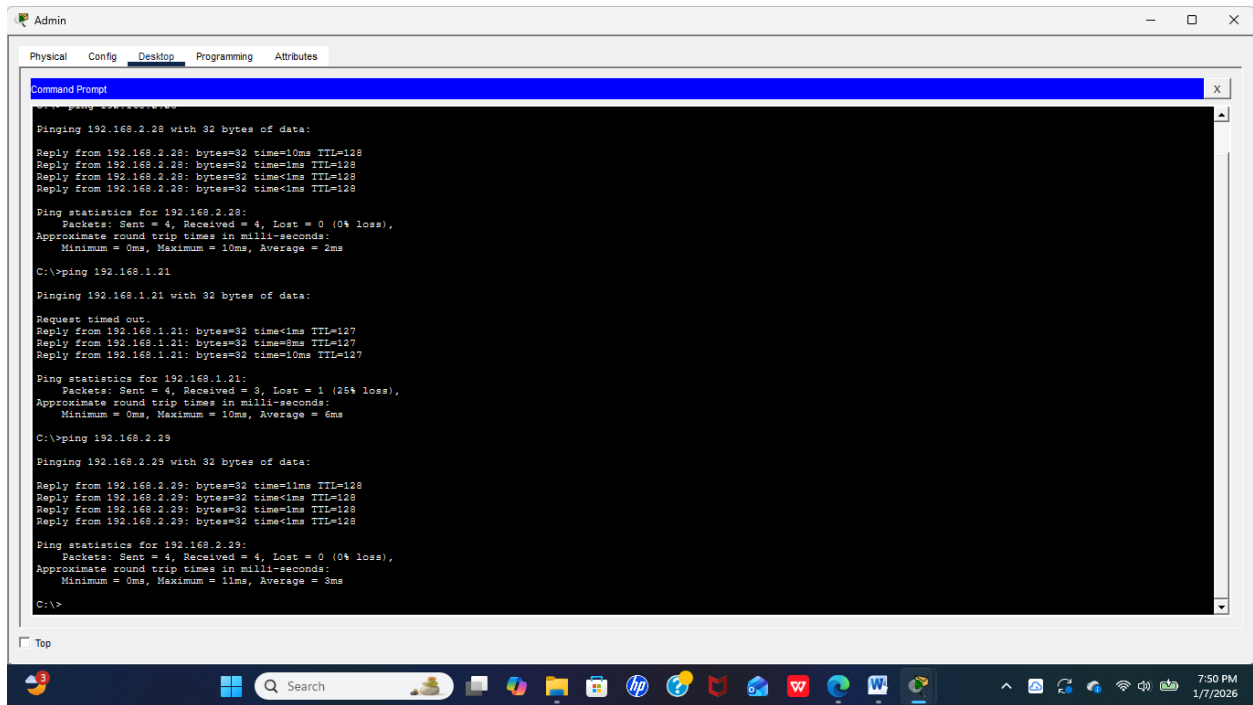
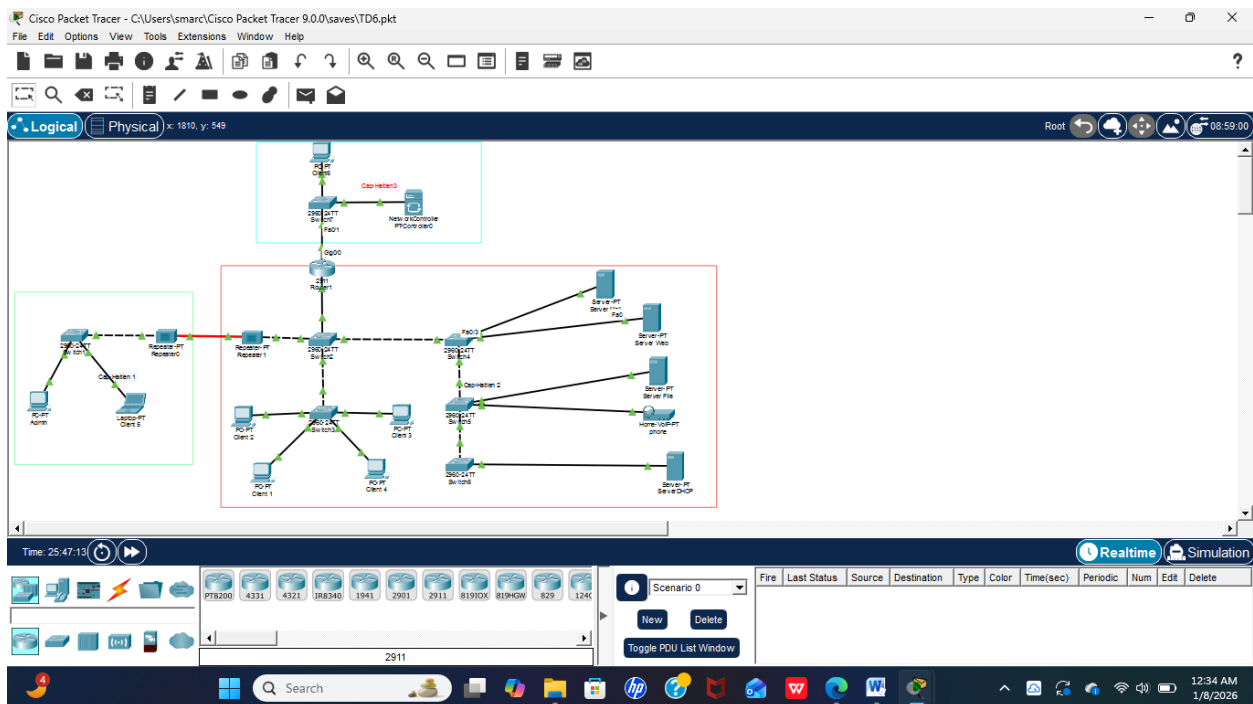


Figure 1 sur Admin j'ai épinglé 3 pcs , il y a connectivité entre les appareils.

4- Reproduisez cette topologie en configurant les services DHCPv6, en utilisant le routeur comme serveur DHCP afin d'attribuer automatiquement les adresses IP aux différents hôtes du réseau.

4.1 Topologie



4.2 Service DHCPv6

Étapes de configuration

Activer IPv6 sur le routeur

Configurer les interfaces

Créer les pools DHCPv6

R1> enable

R1# configure terminal

R1(config)# ipv6 unicast-routing

! Interface Cap1

R1(config)# interface gigabitEthernet 0/0

R1(config-if)# ipv6 address 2001:DB8:1::1/64

R1(config-if)# ipv6 enable

R1(config-if)# no shutdown

! Interface Cap2

R1(config)# interface gigabitEthernet 0/1

R1(config-if)# ipv6 address 2001:DB8:2::1/64

R1(config-if)# ipv6 enable

R1(config-if)# no shutdown

! Pool pour Cap1

R1(config)# ipv6 dhcp pool Cap1

R1(config-dhcpv6)# address prefix 2001:DB8:1::/64

R1(config-dhcpv6)# dns-server 2001:4860:4860::8888

R1(config-dhcpv6)# domain-name reseau-lan1.com

R1(config-dhcpv6)# exit

! Pool pour Cap2

R1(config)# ipv6 dhcp pool Cap2

R1(config-dhcpv6)# address prefix 2001:DB8:2::/64

R1(config-dhcpv6)# dns-server 2001:4860:4860::8888

R1(config-dhcpv6)# domain-name reseau-lan2.com

R1(config-dhcpv6)# exit

```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ipv6 unicast-routing
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ipv6 address 2001:DB8:1::1/64
Router(config-if)#ipv6 enable
Router(config-if)#no shutdown
Router(config-if)#interface gigabitEthernet 0/1
Router(config-if)#ipv6 address 2001:DB8:2::1/64
Router(config-if)#ipv6 enable
Router(config-if)#no sh
Router(config-if)#ipv6 dhcp pool Cap1
Router(config-dhcpv6)#address prefix 2001:DB8:1::/64
Router(config-dhcpv6)#dns-server 2001:4860:4860::8888
Router(config-dhcpv6)#domain-name reseau-lan1.com
Router(config-dhcpv6)#exit
Router(config)#ipv6 dhcp pool Cap2
Router(config-dhcpv6)#address prefix 2001:DB8:2::/64
Router(config-dhcpv6)#dns-server 2001:4860:4860::8888
Router(config-dhcpv6)#domain-name reseau-lan2.com
Router(config-dhcpv6)#exit
Router(config)#
Router(config)#
```

Associer DHCPv6 aux interfaces

! Interface Cap1

R1(config)# interface gigabitEthernet 0/0

R1(config-if)# ipv6 dhcp server Cap1

R1(config-if)# ipv6 nd managed-config-flag

R1(config-if)# exit

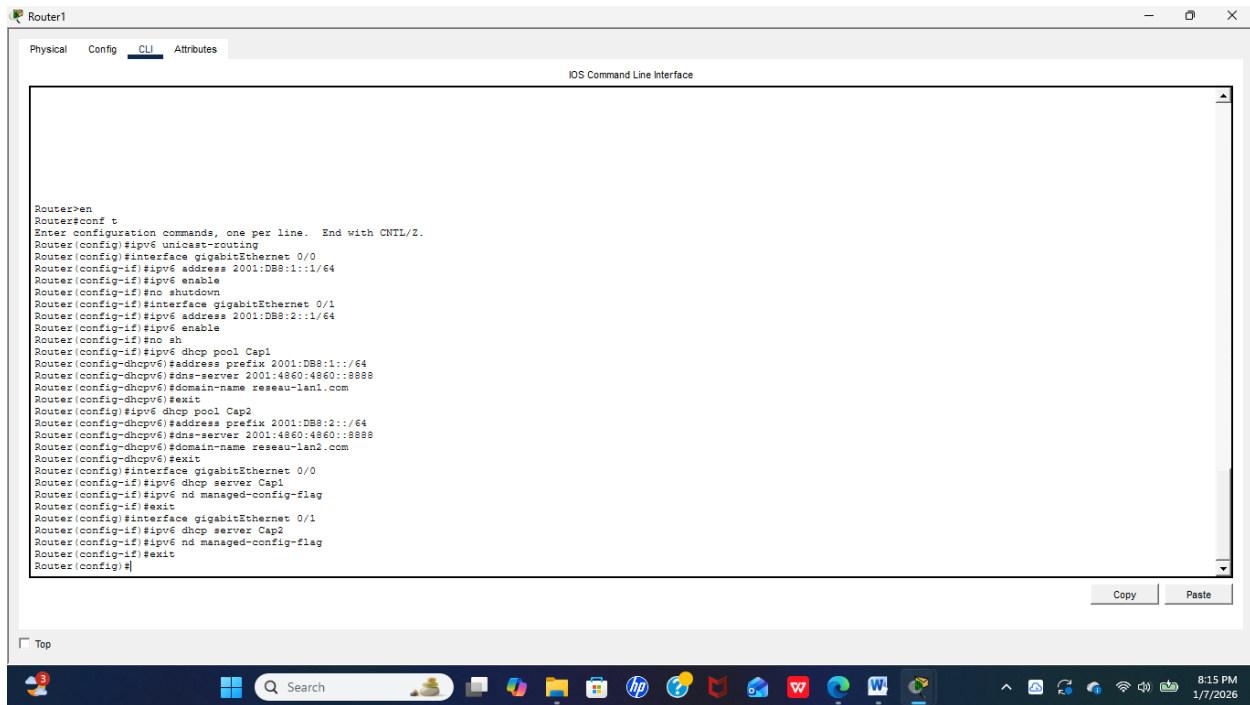
! Interface Cap2

R1(config)# interface gigabitEthernet 0/1

R1(config-if)# ipv6 dhcp server Cap2

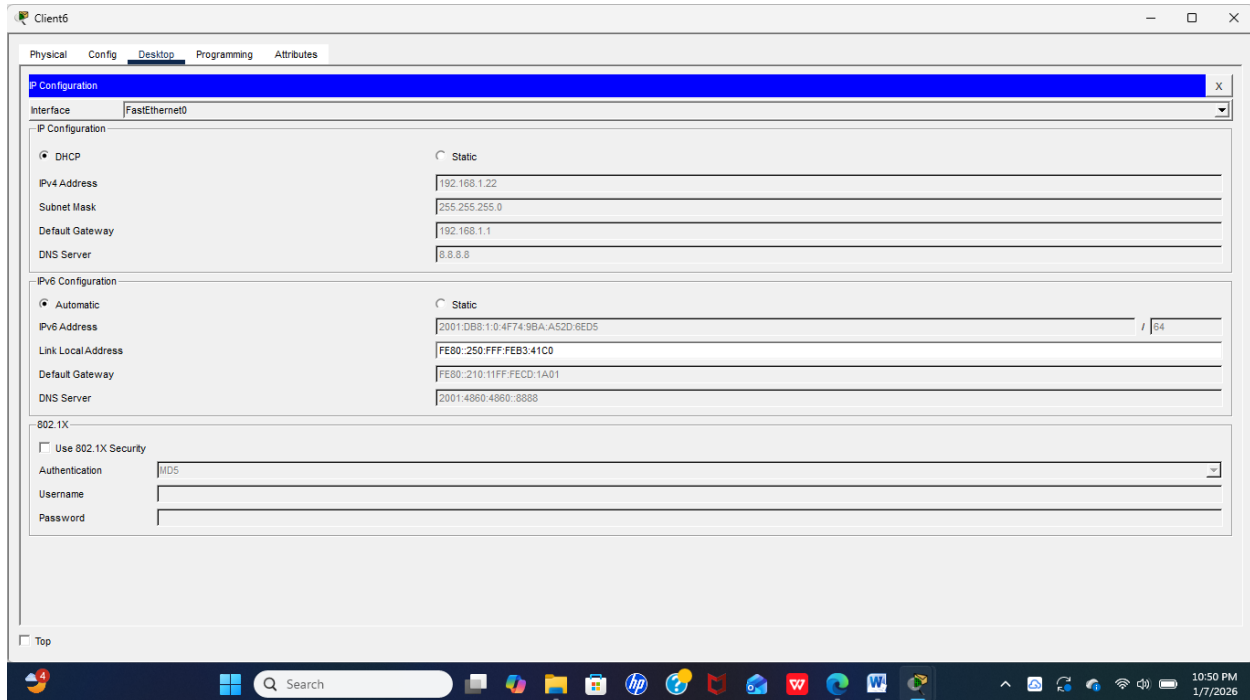
R1(config-if)# ipv6 nd managed-config-flag

R1(config-if)# exit



4.2.1 Configuration des appareil de manière automatique

Client 6



Client 1

Client 1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.2.26

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 8.8.8.8

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address 2001:DB8:1:0:B962:9F0C:2DED:E697 / 64

Link Local Address FE80::202:16FF:FECD:B956

Default Gateway FE80::210:11FF:FECD:1A02

DNS Server 2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

10:47 PM 1/7/2026

Client3

Client 3

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.2.29

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 8.8.8.8

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address 2001:DB8:1:0:A17F:87E9:236D:EC06 / 64

Link Local Address FE80::260:70FF:FE37:D47D

Default Gateway FE80::210:11FF:FECD:1A02

DNS Server 2001:4860:4860::8888

802.1X

☐ Use 802.1X Security

Authentication MD5

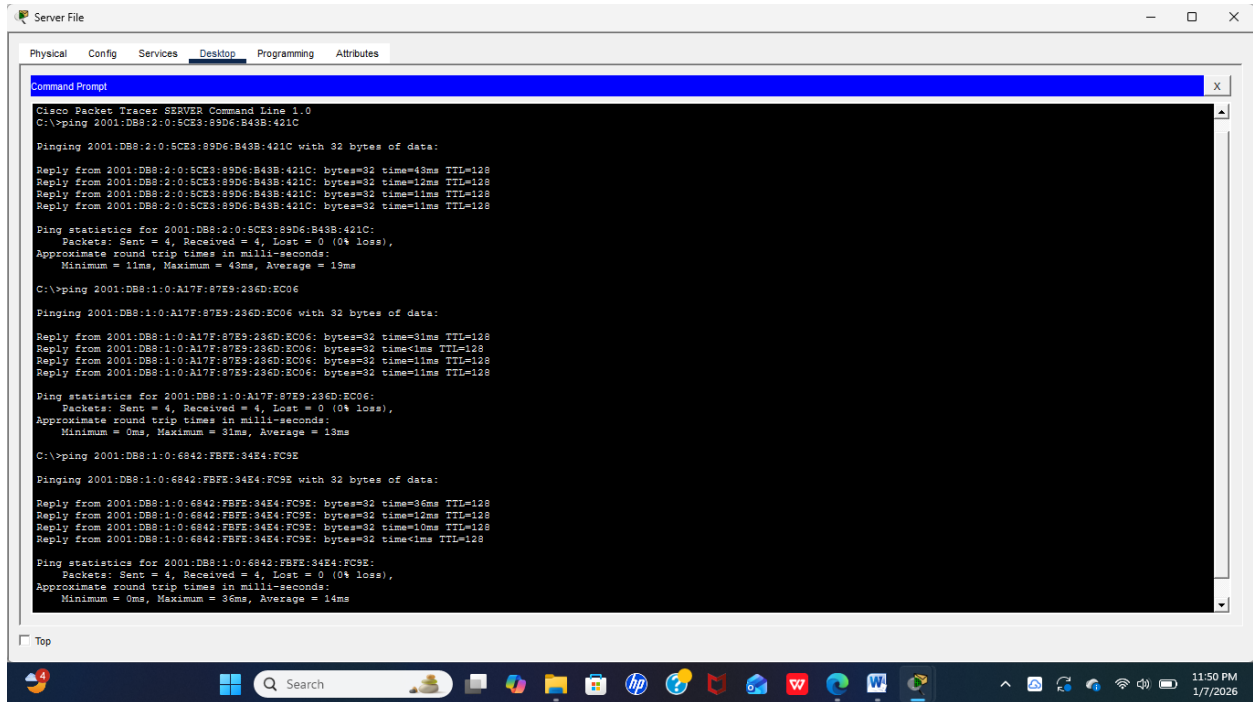
Username

Password

Top

10:46 PM 1/7/2026

4.2.2 Test de connectivité



The screenshot shows a Cisco Packet Tracer window titled "Server File". The "Desktop" tab is selected, displaying a "Command Prompt" window. The Command Prompt shows the following output:

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 2001:DB8:2:0:5CE3:89D6:B43B:421C

Pinging 2001:DB8:2:0:5CE3:89D6:B43B:421C with 32 bytes of data:

Reply from 2001:DB8:2:0:5CE3:89D6:B43B:421C: bytes=32 time=43ms TTL=128
Reply from 2001:DB8:2:0:5CE3:89D6:B43B:421C: bytes=32 time=13ms TTL=128
Reply from 2001:DB8:2:0:5CE3:89D6:B43B:421C: bytes=32 time=11ms TTL=128
Reply from 2001:DB8:2:0:5CE3:89D6:B43B:421C: bytes=32 time=11ms TTL=128

Ping statistics for 2001:DB8:2:0:5CE3:89D6:B43B:421C:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 43ms, Average = 19ms

C:\>ping 2001:DB8:1:0:A17F:87E9:236D:EC06

Pinging 2001:DB8:1:0:A17F:87E9:236D:EC06 with 32 bytes of data:

Reply from 2001:DB8:1:0:A17F:87E9:236D:EC06: bytes=32 time=31ms TTL=128
Reply from 2001:DB8:1:0:A17F:87E9:236D:EC06: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:0:A17F:87E9:236D:EC06: bytes=32 time=11ms TTL=128
Reply from 2001:DB8:1:0:A17F:87E9:236D:EC06: bytes=32 time=1ms TTL=128

Ping statistics for 2001:DB8:1:0:A17F:87E9:236D:EC06:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 31ms, Average = 13ms

C:\>ping 2001:DB8:1:0:6842:FBFE:34E4:FC9E

Pinging 2001:DB8:1:0:6842:FBFE:34E4:FC9E with 32 bytes of data:

Reply from 2001:DB8:1:0:6842:FBFE:34E4:FC9E: bytes=32 time=36ms TTL=128
Reply from 2001:DB8:1:0:6842:FBFE:34E4:FC9E: bytes=32 time=13ms TTL=128
Reply from 2001:DB8:1:0:6842:FBFE:34E4:FC9E: bytes=32 time=10ms TTL=128
Reply from 2001:DB8:1:0:6842:FBFE:34E4:FC9E: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8:1:0:6842:FBFE:34E4:FC9E:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 36ms, Average = 14ms
```

The Command Prompt window is titled "Command Prompt" and has a close button (X) in the top right corner. The window is part of a larger application window titled "Server File" with tabs for "Physical", "Config", "Services", "Desktop", "Programming", and "Attributes". The "Desktop" tab is currently selected. The window is running on a Windows operating system, as indicated by the taskbar at the bottom, which shows the Start button, a search bar, and various application icons. The system clock in the bottom right corner shows the time as 11:50 PM on 1/7/2026.

Figure 2 Sur le Serveur File j'ai épinglé 3 pcs, il y a connectivité entre les appareils.

Conclusion

En conclusion, ce TD a permis de configurer et de tester avec succès les services DHCP et DHCPv6. Les hôtes ont obtenu automatiquement leurs adresses IP, confirmant le bon fonctionnement du réseau et l'importance de l'automatisation de l'adressage.